

Day 10 Task Report

Title: RAG Benchmarking & Optimization (The "Tuning" Task)

TASK-10: Benchmarking and Improving RAG System Performance

Objective

The objective of this task is to evaluate and improve the performance of a Retrieval-Augmented Generation (RAG) system.

The goal is to:

- Measure how well the system answers questions based on the document
 - Identify weaknesses in retrieval
 - Improve performance by tuning system parameters
 - Compare results before and after optimization
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Tools Used

- VS Code
 - Git & GitHub
 - MS Word
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Libraries Used

- **PyPDF2** : To read and extract text from PDF
- **google-genai** : To generate answers using Gemini
- **dotenv** : To manage API keys securely
- **time** : To measure latency
- **re** : For input validation

Concepts Used

1. Benchmarking

Evaluating system performance using predefined test questions.

2. Groundedness Scoring

Measuring how well answers are based on the document (scale 1–5).

3. Chunking Strategy

Splitting text into chunks for retrieval.

4. Retrieval Optimization

Improving how relevant data is selected from the document.

5. RAG Pipeline Improvement

Enhancing accuracy by tuning system parameters.

Code Explanation

The program works in the following steps:

1. Load API key using dotenv
2. Read PDF using PyPDF2
3. Extract and chunk text into smaller parts
4. Define 5 fixed test questions
5. Run system in two phases:
 - Before tuning (chunk size = 200)
 - After tuning (chunk size = 600 with overlap)
6. For each question:
 - Retrieve relevant chunk
 - Generate answer using Gemini

- Assign groundedness score (1–5) manually
7. Store scores for both runs
 8. Display comparison table
 9. Calculate average performance
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Benchmark Questions

1. I'm seeing Error XV-505 in my terminal. What is the root cause and how do I fix it?
 2. My system is extremely laggy. Which shards should I be worried about and what is the specific command to reset them?
 3. Can I bypass the Theta-Sync requirement to install this on my old i7 laptop?
 4. How does the OS handle user login now that passwords and 2FA are deprecated?
 5. I have a MAC address that was black-holed by the Wraith Firewall. Can I unblock it using the Galactic-Relay?
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Errors Faced

1. Poor Retrieval Accuracy

Problem:

System failed to retrieve correct information for several queries. Because Small chunk size (200) split important information.

Solution:

Increased chunk size to 600.

2. Context Splitting Issue

Problem:

Important information was divided across multiple chunks.

Solution:

Implemented overlapping chunking:

for i in range(0, len(text), size - overlap):

3. Weak Keyword Matching**Problem:**

Simple keyword matching failed to capture meaning.

Solution:

Improved scoring logic by weighting matches.

4. "I DON'T KNOW" Responses**Problem:**

Model failed to answer even when data existed.

Cause:

Incorrect chunk retrieval.

Solution:

Improved chunking and retrieval logic.

Comparison Analysis

- Before tuning, system performance was low due to weak retrieval
- After tuning, answers became more accurate and context-based
- Increasing chunk size improved context availability
- Overlapping chunks prevented information loss

Output

```
(.venv) PS D:\Qualtech\Daily Tasks\Day10> py -3.10 sys-improvement.py
Loading PDF...

--- BEFORE TUNING (chunk size = 200) ---

Q: I'm seeing Error XV-505 in my terminal. What is the root cause and how do I fix it?
Answer: I DON'T KNOW
Enter groundedness score (1-5): 1

Q: My system is extremely laggy. Which shards should I be worried about and what is the specific command to reset them?
Answer: I DON'T KNOW
Enter groundedness score (1-5): 1

Q: Can I bypass the Theta-Sync requirement to install this on my old i7 laptop?
Answer: Yes, you can bypass it using 'sudo override --bypass-shards', but it is risky and only for emergency use.
Enter groundedness score (1-5): 5

Q: How does the OS handle user login now that passwords and 2FA are deprecated?
Answer: The system uses Biometric-Frequency Identification (BFI), based on heartbeat resonance.
Enter groundedness score (1-5): 5

Q: I have a MAC address that was black-holed by the Wraith Firewall. Can I unblock it using the Galactic-Relay?
Answer: I DON'T KNOW
Enter groundedness score (1-5): 1
```

```
--- AFTER TUNING (chunk size = 600) ---

Q: I'm seeing Error XV-505 in my terminal. What is the root cause and how do I fix it?
Answer: Error XV-505 is caused by incorrect VORTEX_PATH mapping. Fix it by setting VORTEX_PATH to /pulse/bin.
Enter groundedness score (1-5): 5

Q: My system is extremely laggy. Which shards should I be worried about and what is the specific command to reset them?
Answer: System lag is related to Shard-07 and Shard-04. Use command: vortex --realign-pulse.
Enter groundedness score (1-5): 5

Q: Can I bypass the Theta-Sync requirement to install this on my old i7 laptop?
Answer: Yes, bypass using 'sudo override --bypass-shards', but it is risky.
Enter groundedness score (1-5): 4

Q: How does the OS handle user login now that passwords and 2FA are deprecated?
Answer: Login uses Biometric-Frequency Identification (BFI) using heartbeat resonance.
Enter groundedness score (1-5): 5

Q: I have a MAC address that was black-holed by the Wraith Firewall. Can I unblock it using the Galactic-Relay?
Answer: No, remote unblocking is not possible. Requires physical hardware key-turn.
Enter groundedness score (1-5): 5
```

```
==== FINAL COMPARISON ====
Q.No      Before  After
1         1      5
2         1      5
3         5      4
4         5      5
5         1      5

Average Before: 2.6
Average After : 4.8
```